Two-use additivity of output ergotropy for generalized amplitude damping

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We consider two noninteracting unit-gap qubits subjected independently to the same generalized amplitude-damping channel. At a fixed upper bound on the mean input energy, we optimize the output ergotropy over all complex density operators and allow a global work-extraction unitary. A previous numerical study found no two-use advantage and conjectured additivity, but its reduction to real amplitudes does not cover the full input set. We prove exactly that the unrestricted two-use optimum equals twice the one-use optimum for every damping parameter $\gamma \in [0, 1]$, equilibrium excitation $\eta \in [0, 1/2]$, and per-cell budget $e \in [0, 1]$. The proof first reduces the energy-ball optimization to pure inputs while retaining the one invariant two-qubit phase. It then converts ergotropy to an output spectral-diameter bound. Exact formulas for the second and fourth elementary spectral coefficients reduce the problem to a quadratic inequality, which is closed by a sign-definite multiplier identity. Thus entangling two-cell encodings cannot increase maximal extractable work in this model. The result concerns exactly two parallel uses and does not establish additivity for arbitrary numbers of cells.

I. INTRODUCTION

Ergotropy quantifies the work extractable from a finite quantum state by a cyclic unitary operation [1]. For quantum batteries, environmental noise motivates an encoding problem: given a constraint on the energy stored before the noise, how much ergotropy can remain afterward [2, 3]? Joint encodings and global extraction can in principle make the multi-cell optimization superadditive, which is why noisy work capacitances distinguish local, separable, and unrestricted resources [4, 5].

The qubit generalized amplitude-damping channel (GADC) is the standard finite-temperature extension of amplitude damping [6]. For this channel, Tirone *et al.* derived the one-use maximal output ergotropy and performed a nonexhaustive two-use search [7]. They found no violation and conjectured additivity for arbitrary numbers of uses. Reference [7] left the unrestricted two-use question unresolved: a finite search is not a universal certificate, and the assertion used there that all four phases of a pure two-qubit input can be removed is false. Three phases are removable by a global phase and local energy rotations, but one combination remains invariant and changes the output spectrum.

Here we prove the full two-use statement without assuming real amplitudes. The result covers all mixed and pure two-qubit inputs satisfying an *energy-ball* constraint and allows global output passive-state rearrangement. The proof uses only finite-dimensional exact algebra. Its main ingredients are: (i) an extreme-point reduction of the input ball; (ii) an identity bounding a four-level spectral diameter through the second and fourth elementary coefficients; (iii) exact phase-dependent coefficient formulas; and (iv) a polynomial multiplier whose remainder is a sum of nonnegative monomials. A recent multilevel finite-temperature construction also leaves entangled-input advantages as a future direction [8]; our result is restricted to the qubit GADC and two parallel uses.

II. MODEL, ONE-USE OPTIMUM, AND RESULT

Each cell has energy basis $(|0\rangle, |1\rangle)$ and Hamiltonian $h = |1\rangle\langle 1|$. For $n = 1, 2$ noninteracting cells,

$$H_n = \sum_{j=1}^{n} h_j. \tag{1}$$

The energy gap is set to one; restoring a gap Δ multiplies every energy and ergotropy below by Δ. The local channel is

$$\Phi_{\gamma,\eta}(\rho) = \sum_{\nu=0}^{3} K_\nu \rho K_\nu^\dagger, \tag{2}$$

with

$$K_0 = \sqrt{1-\eta}\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \sqrt{(1-\eta)\gamma}\,|0\rangle\langle 1|,$$
$$K_2 = \sqrt{\eta}\begin{pmatrix} \sqrt{1-\gamma} & 0 \\ 0 & 1 \end{pmatrix}, \qquad K_3 = \sqrt{\eta\gamma}\,|1\rangle\langle 0|. \tag{3}$$

We take $0 \le \gamma \le 1$ and $0 \le \eta \le 1/2$. The parameter η is the equilibrium excited-state population. At a fixed positive total transition rate, finite time corresponds to $\gamma < 1$; $\gamma = 1$ is included as the limiting replacement map. In matrix form,

$$\Phi_{\gamma,\eta}(\rho) = \begin{pmatrix} 1-\gamma\eta-(1-\gamma)\rho_{11} & \sqrt{1-\gamma}\rho_{01} \\ \sqrt{1-\gamma}\rho_{10} & \gamma\eta+(1-\gamma)\rho_{11} \end{pmatrix}. \tag{4}$$

For a state ω and Hamiltonian H, its ergotropy is

$$\mathcal{W}(\omega, H) = \mathrm{Tr}(\omega H) - \min_U \mathrm{Tr}(U\omega U^\dagger H). \tag{5}$$

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For a per-cell input-energy budget $e \in [0,1]$, define

$$\mathcal{D}_n(e) = \{\rho \succeq 0 : \operatorname{Tr}\rho = 1,\ \operatorname{Tr}(\rho H_n) \le ne\}, \quad (6)$$

$$W_n^{\gamma,\eta}(e) = \max_{\rho\in\mathcal{D}_n(e)} \mathcal{W}\big(\Phi_{\gamma,\eta}^{\otimes n}(\rho), H_n\big). \quad (7)$$

The channel gives the exact energy law

$$\operatorname{Tr}[\Phi_{\gamma,\eta}^{\otimes n}(\rho)H_n] = (1-\gamma)\operatorname{Tr}(\rho H_n) + n\gamma\eta. \quad (8)$$

We will repeatedly use

$$u = \gamma\eta, \qquad d = \gamma(1-\eta), \qquad q = 1-\gamma. \quad (9)$$

Thus $0 \le u \le d$, $u+d \le 1$, and $q = 1-u-d$. For a pure one-cell input with excitation x, let

$$r(x) = \sqrt{(1-2u)^2 + 4\gamma q(2\eta - x)x}. \quad (10)$$

The one-use shell ergotropy and ball optimum are

$$w(x) = qx + u - \frac{1}{2} + \frac{1}{2}r(x), \quad (11)$$

$$W_1^{\gamma,\eta}(e) = w(x_e), \qquad x_e = \min\{e, x_*\}, \quad (12)$$

where, for $\gamma > 0$,

$$x_* = \min\left\{1, \eta + \frac{\sqrt{1-4\gamma\eta(1-\eta)}}{2\sqrt{\gamma}}\right\}, \quad (13)$$

and $x_* = 1$ at $\gamma = 0$. Equation (12) is the published one-use result [7].

Our main result is the following.

$$\boxed{W_2^{\gamma,\eta}(e) = 2W_1^{\gamma,\eta}(e) \quad \text{for all } \gamma\in[0,1],\ \eta\in[0,1/2],\ e\in[0,1].} \quad (14)$$

The optimization on the left of Eq. (14) is over the full complex density-operator ball, and its extraction unitary is global. No ancilla, interaction, correlated noise, measurement, feedback, postselection, or preparation-cost term is included.

III. REDUCTION TO A SPECTRAL-DIAMETER BOUND

A. Mixed inputs and the invariant phase

For fixed channel parameters, the map

$$F(\rho) = \mathcal{W}(\Phi_{\gamma,\eta}^{\otimes 2}(\rho), H_2) \quad (15)$$

is convex: Eq. (5) is a pointwise maximum of linear functions of its state argument, and the channel is linear. A continuous convex function on the compact set $\mathcal{D}_2(e)$ has a maximizer at an extreme point. Every such extreme point has rank one. Indeed, if a feasible state has rank $r \ge 2$, Hermitian perturbations on its support form a real vector space of dimension $r^2 \ge 4$. One can choose a nonzero supported perturbation X with $\operatorname{Tr} X = 0$ and, when the energy inequality is active, $\operatorname{Tr}(XH_2) = 0$. For sufficiently small $\epsilon > 0$, both $\rho \pm \epsilon X$ remain positive and feasible, contradicting extremality. If the inequality is slack, ϵ can also be chosen to preserve the slack. Hence it suffices to bound pure inputs.

Energy-phase covariance of the GADC, together with a global phase, removes three input phases. Up to zero-amplitude limits, an arbitrary relevant pure state can therefore be written

$$|\psi\rangle = \sqrt{A}|00\rangle + \sqrt{B}|01\rangle + \sqrt{C}|10\rangle + e^{i\delta}\sqrt{P}|11\rangle, \quad (16)$$

where $A, B, C, P \ge 0$ and $A+B+C+P = 1$. The actual per-cell energy is

$$e_\psi = \frac{B + C + 2P}{2}. \quad (17)$$

The phase δ cannot generally be removed. For example, at $(\gamma,\eta) = (1/2, 1/4)$ the equal-population states with amplitudes $(1,1,1,1)/2$ and $(1,1,1,-1)/2$ have output purities $625/1024$ and $417/1024$, respectively. Figure 1(b) shows the continuous phase dependence. Our proof retains δ throughout.

B. Ergotropy through the output diameter

Let the two-use output eigenvalues be $\lambda_1 \ge \lambda_2 \ge \lambda_3 \ge \lambda_4$. Since the two-cell energy levels are $(0,1,1,2)$, the passive energy is

$$\lambda_2 + \lambda_3 + 2\lambda_4 = 1 - (\lambda_1 - \lambda_4). \quad (18)$$

Writing $D = \lambda_1 - \lambda_4$ and setting $e = e_\psi$, Eq. (8) gives the exact output ergotropy

$$\mathcal{W}_2(\psi) = 2(qe + u) - 1 + D. \quad (19)$$

Comparison with Eq. (12) shows that the desired upper bound is

$$D \le r(x_e) - 2q(e - x_e). \quad (20)$$

We will prove the stronger shell inequality $D \le r(e)$. It immediately implies Eq. (20) when $x_e = e$. When the

one-use optimizer saturates, $x_e = x_* < e$, the concavity of r and the stationarity condition $r'(x_*) = -2q$ give

$$r(e) \le r(x_*) - 2q(e - x_*), \tag{21}$$

which again implies Eq. (20).

IV. EXACT COEFFICIENT PROOF

A. A universal four-level inequality

For any four-level density spectrum, define

$$E_2 = \sum_{i<j} \lambda_i \lambda_j, \qquad E_4 = \prod_{i=1}^{4} \lambda_i. \tag{22}$$

Direct expansion gives the sum-of-squares identity

$$1 - D^2 - 2E_2 - 4\sqrt{E_4} = (\lambda_2 - \lambda_3)^2 + 2\left(\sqrt{\lambda_1\lambda_4} - \sqrt{\lambda_2\lambda_3}\right)^2 \ge 0. \tag{23}$$

For a pure one-cell input of excitation e, the determinant of its GADC output is

$$d_e = \alpha - 2uqe + (u+d)qe^2, \qquad \alpha = u(1-u), \tag{24}$$

and Eqs. (10) and (24) obey

$$1 - r(e)^2 = 4d_e. \tag{25}$$

Consequently, Eqs. (23) and (25) prove $D \le r(e)$ once we establish

$$E_2 + 2\sqrt{E_4} \ge 2d_e. \tag{26}$$

B. Phase-dependent output coefficients

For the state in Eq. (16), put

$$S = B + C = 2(e - P), \qquad v = BC, \\ y = \sqrt{APv}\cos\delta. \tag{27}$$

Thus $A + S + P = 1$, $0 \le v \le S^2/4$, and $|y| \le \sqrt{APv}$. Define

$$\beta = d(1-d), \quad c = d - u, \quad \tau = u + d - 2ud, \\ R_u = 1 - 2u + 2u^2. \tag{28}$$

Exact collection of the full phase-dependent output in Appendix A gives

$$E_2 = 2\alpha(1-\alpha) - 2c^2q^2P^2 + 4ucq^2PS + 2cqR_uP + (u+d)qR_uS^2 - 2uqR_uS - 4q\tau y. \tag{29}$$

In particular, E_2 is independent of v at fixed (A, S, P, y).

Let D_0 denote the determinant at $v = y = 0$, explicitly given in Appendix A, and set

$$K = AS\alpha + SP\beta + 4APud, \\ X_0 = A\alpha + Sud + P\beta, \\ X_1 = A\alpha + S(1-u)(1-d) + P\beta. \tag{30}$$

The full determinant is affine in the split and phase invariants:

$$E_4 = D_0 + 4udq^2Kv - 8udqX_0X_1y. \tag{31}$$

The coefficient of v is nonnegative. At fixed (A, S, P, y) and $AP > 0$, physical admissibility implies

$$\frac{y^2}{AP} \le v \le \frac{S^2}{4}. \tag{32}$$

The lower endpoint is physically realized by $\delta = 0$ or π and a suitable B, C split. Therefore Eq. (26) is hardest at

$$v = \frac{y^2}{AP}, \qquad -\frac{S\sqrt{AP}}{2} \le y \le \frac{S\sqrt{AP}}{2}, \tag{33}$$

with boundary cases defined by continuity. Denote the resulting determinant by

$$E_{4,\min} = D_0 + \frac{4udq^2K}{AP}y^2 - 8udqX_0X_1y. \tag{34}$$

C. Multiplier certificate

Write $g = u + d$ so that $q = 1 - g$, and introduce

$$R = A\alpha + P\beta, \quad X = X_0X_1, \quad K = SR + 4APud, \\ J = X_0^2 + udq(S^2 - 4AP). \tag{35}$$

The channel identities $(1-u)(1-d) = q + ud$ and $\alpha\beta = ud(q + ud)$ imply

$$X = qK + J, \qquad KD_0 - 4APudX^2 = SRJ^2. \tag{36}$$

These are polynomial identities; the second avoids any division at a boundary. They also show that the worst-split determinant has an exact sum-of-squares representation away from removable zero denominators.

Let

$$T = d_e - \frac{E_2}{2}, \qquad Q = E_{4,\min} - T^2, \tag{37}$$

and define the nonnegative interval polynomials

$$G = \frac{S^2}{4} + AP - 2y, \qquad H = \frac{APS^2}{4} - y^2. \tag{38}$$

Equation (33) gives $G, H \ge 0$, and

$$APG - H = (AP - y)^2, \qquad H \le APG. \tag{39}$$

Finally, set

$$\kappa = g(g-4ud), \qquad F = g(2-g) - 4udq \tag{40}$$

and

$$\mathcal{R} = g^2\alpha^2A^2 + 2ud\alpha FAS + 2u^2d^2g^2AP + \frac{udg^2(q+2ud)}{2}S^2 + 2ud\beta FSP + g^2\beta^2P^2. \tag{41}$$

Exact expansion using $A+S+P=1$ yields the central multiplier identity

$$AP\tau Q = APqG\big(\kappa T + \mathcal{R}\big) - 4Sudq^2\tau RH. \tag{42}$$

We now use Eq. (42) only in the sector $T>0$; when $T\le 0$, Eq. (26) is automatic. Since $4ud \le g^2 \le g$, one has $\kappa \ge 0$. Moreover,

$$F - 2q\tau = g^2. \tag{43}$$

It follows that

$$\begin{aligned}\mathcal{R}_0 &:= \mathcal{R} - 4Sudq\tau R \\ &= g^2\alpha^2A^2 + 2ud\alpha g^2AS + 2u^2d^2g^2AP \\ &\quad + \frac{udg^2(q+2ud)}{2}S^2 + 2ud\beta g^2SP + g^2\beta^2P^2 \ge 0.\end{aligned} \tag{44}$$

Using $H \le APG$ in Eq. (42) therefore gives

$$AP\tau Q \ge APqG\big(\kappa T + \mathcal{R}_0\big) \ge 0. \tag{45}$$

For $g>0$ one has $\tau>0$, so Eq. (45) proves $Q\ge 0$ for $AP>0$. If $AP=0$, then $y=0$, and the result follows by continuity of the same polynomial identities. The cases $g=0$ and $q=0$ are treated directly below. Thus, whenever $T>0$, $\sqrt{E_{4,\min}} \ge T$; together with the automatic $T\le 0$ sector, this proves Eq. (26). Because E_4 is nondecreasing in the discarded variable v, the conclusion holds for every invariant phase and population split.

Equations (23), (25), and (26) now give $D \le r(e)$. The tangent step (21) completes the upper bound for every pure input shell, the extreme-point argument covers mixed inputs, and monotonicity of the ball optimum covers states with actual energy smaller than the available budget.

For the reverse inequality, take a one-use optimizer ρ_* and use $\rho_*^{\otimes 2}$. If the eigenvalues of $\sigma_* = \Phi_{\gamma,\eta}(\rho_*)$ are $\lambda_+ \ge \lambda_-$, the passive energy of $\sigma_*^{\otimes 2}$ is $2\lambda_-$, twice the passive energy of σ_*. Hence

$$\mathcal{W}(\sigma_*^{\otimes 2}, H_2) = 2\mathcal{W}(\sigma_*, h), \tag{46}$$

which proves the matching product-code lower bound and Eq. (14).

At $\gamma = 0$ the channel is the identity, so the maximum ergotropy equals the available energy for one or two cells. At $\gamma = 1$ it replaces every input by $\mathrm{diag}(1-\eta,\eta)^{\otimes n}$, which is passive for $\eta \le 1/2$; both sides of Eq. (14) vanish. These direct arguments include the closed-domain endpoints without interpreting $\gamma = 1$ as finite-time fixed-rate evolution.

V. VERIFICATION, IMPLICATIONS, AND LIMITATIONS

The identities in Eqs. (29), (31), (36), and (42) were expanded and reduced to zero in exact symbolic arithmetic. Independently transcribed channel implementations—one summing the sixteen product Kraus operators and one tensoring the Liouville action of Eq. (4) with an explicit index permutation—agree on the full two-qubit operator basis. They reproduce the one-use formula and product-code equality. Floating-point searches over complex pure and mixed inputs provide diagnostics but are not used as evidence for the universal inequality; the theorem rests on the exact sign certificate above. The substantive AI assistance used in the research and verification workflow is identified in the Acknowledgments; all machine-produced algebra entering the result was subjected to the exact and independent checks just described.

Operationally, Eq. (14) says that an entangling two-cell encoding cannot improve maximal output ergotropy at a fixed per-cell input budget under identical, independent GADC noise. It does not say that every optimal input is separable, nor does it price the preparation of an entangled state. More importantly, the proof is specific to two uses: neither the four-level spectral identity nor the coefficient reduction establishes the arbitrary-n conjecture in Ref. [7]. The conclusion also does not cover interacting cells, correlated noise, ancillas, local-only extraction, feedback, postselection, or intermediate control.

DATA AVAILABILITY

The exact symbolic checkers, the independent Kraus and Liouville channel implementations, captured verification logs, and the source data and script for Fig. 1 accompany this local manuscript package. No experimental data were used. A permanent public repository identifier must be chosen by the author before submission.

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OpenAI Codex using the gpt-5.6-sol model with the Ultra reasoning setting was used substantively for scientific reasoning, symbolic-code development, independent computational checks, literature organization, and manuscript drafting. The research scope and quantified claim were explicitly frozen by the author. AI-produced algebra was checked by exact symbolic identities, independent Kraus/Liouville implementations, and a separate AI referee replay. This local draft still requires complete human scientific and textual review; before any submission the author must review and refine the work and assume full responsibility for its accuracy.

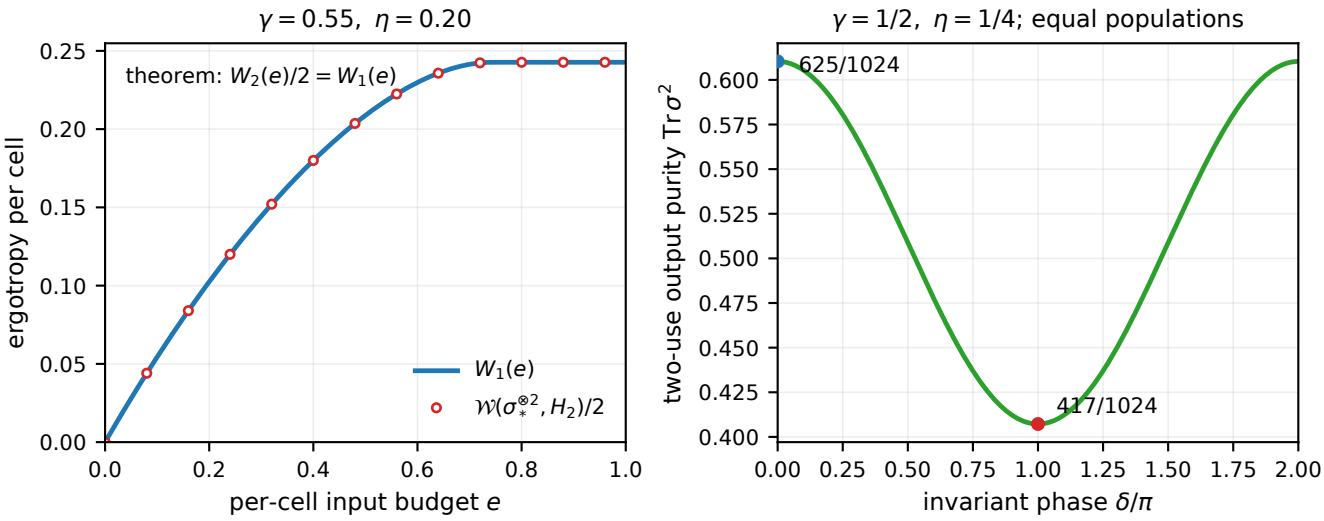


FIG. 1. (a) The one-use optimum from Eq. (12) (solid curve) and half the ergotropy of the corresponding two-copy product output, evaluated by a direct Kraus sum (circles). The theorem promotes the product value to the unrestricted two-use optimum $W_2(e)/2$. The optimizer stops using additional budget when $e > x_*$. (b) Output purity for equal input populations as the invariant phase varies. The exact values at $\delta = 0$ and π show why a real-amplitude or phase-independent reduction cannot certify the theorem. The plotted data and generation script accompany the manuscript. OpenAI Codex (gpt-5.6-sol, Ultra reasoning setting) assisted with the plotting code and figure generation; deterministic replay reproduced both CSV files byte-for-byte and the 180-dpi figure raster pixel-for-pixel.

Appendix A: Full phase-dependent output and the reference determinant

For the state in Eq. (16), the diagonal entries of $\sigma = \Phi_{\gamma,\eta}^{\otimes 2}(|\psi\rangle\langle\psi|)$ in the ordered basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$ are

$$\begin{aligned} z_0 &= (1-u)^2 A + (1-u)dS + d^2 P, \\ z_1 &= u(1-u)A + (1-u)(1-d)B + udC + d(1-d)P, \\ z_2 &= u(1-u)A + udB + (1-u)(1-d)C + d(1-d)P, \\ z_3 &= u^2 A + u(1-d)S + (1-d)^2 P. \end{aligned} \tag{A1}$$

Its upper-triangular off-diagonal entries are

$$\begin{aligned} \sigma_{01} &= \sqrt{q}\left[(1-u)\sqrt{AB} + d\sqrt{CP}e^{-i\delta}\right], \\ \sigma_{02} &= \sqrt{q}\left[(1-u)\sqrt{AC} + d\sqrt{BP}e^{-i\delta}\right], \\ \sigma_{03} &= q\sqrt{AP}e^{-i\delta}, \qquad \sigma_{12} = q\sqrt{BC}, \\ \sigma_{13} &= \sqrt{q}\left[u\sqrt{AC} + (1-d)\sqrt{BP}e^{-i\delta}\right], \\ \sigma_{23} &= \sqrt{q}\left[u\sqrt{AB} + (1-d)\sqrt{CP}e^{-i\delta}\right]. \end{aligned} \tag{A2}$$

The lower-triangular entries are fixed by Hermiticity.

To specify D_0 in Eq. (31), choose $B = S$, $C = 0$ and define

$$\begin{aligned} z_1^{(0)} &= u(1-u)A + (1-u)(1-d)S + d(1-d)P, \\ z_2^{(0)} &= u(1-u)A + udS + d(1-d)P, \end{aligned} \tag{A3}$$

and

$$\begin{aligned} a_{01} &= \sqrt{qAS}(1-u), \quad a_{02} = \sqrt{qSP}\,d, \quad a_{03} = q\sqrt{AP}, \\ a_{13} &= \sqrt{qSP}(1-d), \qquad a_{23} = \sqrt{qAS}\,u. \end{aligned} \tag{A4}$$

Then the explicitly polynomial expression (each displayed square-root product has integer powers) is

$$\begin{aligned} D_0 = {} & z_0 z_1^{(0)} z_2^{(0)} z_3 - a_{01}^2 z_2^{(0)} z_3 - a_{02}^2 z_1^{(0)} z_3 \\ & - a_{03}^2 z_1^{(0)} z_2^{(0)} - a_{13}^2 z_0 z_2^{(0)} - a_{23}^2 z_0 z_1^{(0)} \\ & + a_{01}^2 a_{23}^2 + a_{02}^2 a_{13}^2 \\ & - 2a_{01}a_{02}a_{13}a_{23} + 2a_{01}a_{03}a_{13} z_2^{(0)} \\ & + 2a_{02}a_{03}a_{23} z_1^{(0)}. \end{aligned} \tag{A5}$$

Expanding Eqs. (A1)–(A2) gives Eqs. (29) and (31); no phase-independent ansatz is used.

Appendix B: Exact-check map

The accompanying checker first constructs the entries in Eqs. (A1)–(A2) with a Laurent variable for $e^{i\delta}$ and verifies Eq. (31) by exact fraction-free elimination. A second checker expands the following identities independently:

1. $X = qK + J$ and $KD_0 - 4APudX^2 = SRJ^2$;
2. the multiplier identity (42);
3. $F - 2q\tau = g^2$ and the nonnegative remainder (44);
4. the interval identity (39), the one-use identity (25), and the spectral identity (23).

All of these checks are exact polynomial reductions. The numerical routines then compare the two channel representations and evaluate boundary-realizing states; they

are retained to catch transcription and basis-order errors, not to replace the proof.

[1] A. E. Allahverdyan, R. Balian, and T. M. Nieuwenhuizen, Maximal work extraction from finite quantum systems, Europhysics Letters **67**, 565 (2004).

[2] R. Alicki and M. Fannes, Entanglement boost for extractable work from ensembles of quantum batteries, Physical Review E **87**, 042123 (2013).

[3] S. Tirone, R. Salvia, and V. Giovannetti, Quantum energy lines and the optimal output ergotropy problem, Physical Review Letters **127**, 210601 (2021).

[4] S. Tirone, R. Salvia, S. Chessa, and V. Giovannetti, Work extraction processes from noisy quantum batteries: The role of nonlocal resources, Physical Review Letters **131**, 060402 (2023).

[5] S. Tirone, R. Salvia, S. Chessa, and V. Giovannetti, Quantum work capacitances: Ultimate limits for energy extraction on noisy quantum batteries, SciPost Physics **17**, 041 (2024).

[6] S. Khatri, K. Sharma, and M. M. Wilde, Information-theoretic aspects of the generalized amplitude-damping channel, Physical Review A **102**, 012401 (2020).

[7] S. Tirone, R. Salvia, S. Chessa, and V. Giovannetti, Quantum work extraction efficiency for noisy quantum batteries: The role of coherence, Physical Review A **111**, 012204 (2025), arXiv:2305.16803 [quant-ph].

[8] V. Vetrano, V. Giovannetti, and V. Cavina, Generalized multilevel amplitude damping channels and their thermodynamic performances (2026), arXiv:2605.27369v1, arXiv:2605.27369 [quant-ph].